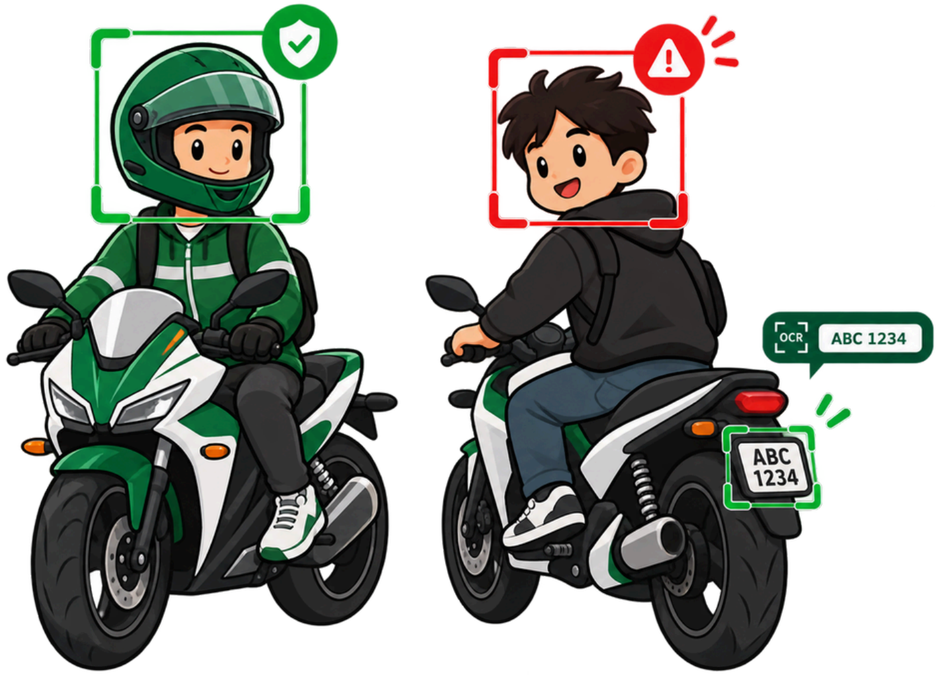
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Motorcycle Helmet Violation Detection

Deep Learning-based traffic safety system using YOLOv8 + EasyOCR



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Course: Deep Learning
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School: Kirirom Institute of Technology
Date: 25 September 2026

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Project Overview

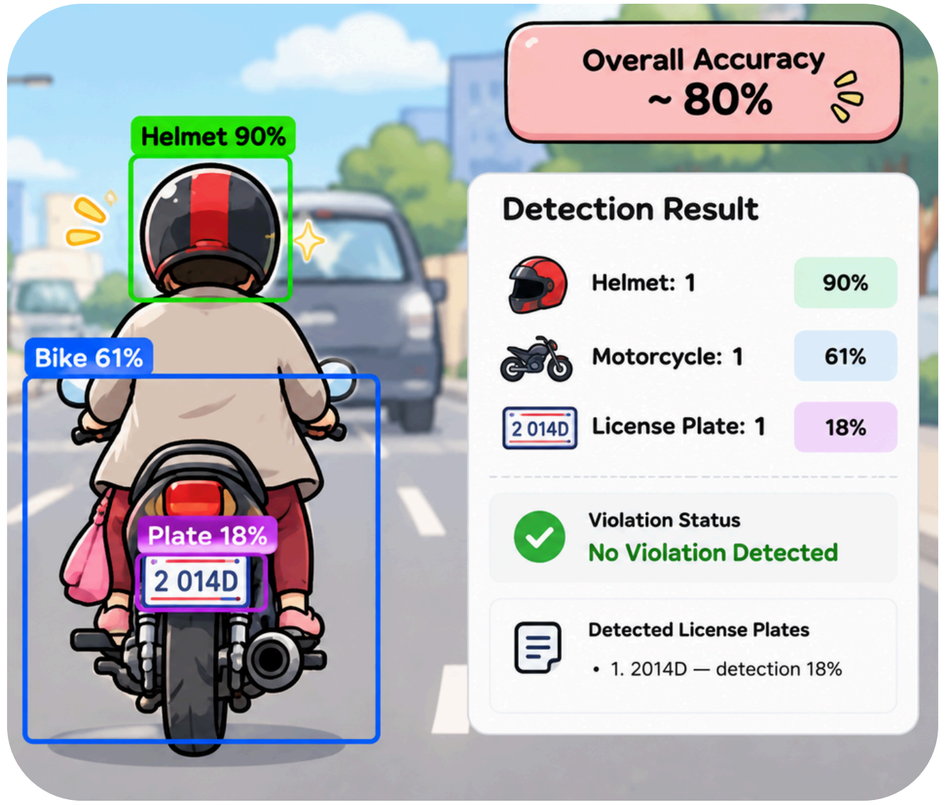
Main focus: helmet and no-helmet violation detection

Main Feature

- Helmet / No-Helmet Detection
- Detects helmet and no-helmet riders
- Identifies helmet violations
- Shows bounding boxes and confidence
- Runs in a Gradio web app

Supporting Features

- Motorcycle Detection
- Helps locate motorcycles and associate riders.
- License Plate Detection + OCR
- Detects plate regions and reads text using EasyOCR.



Problem Statement

Manual helmet monitoring is difficult in real traffic scenes

- Some riders do not wear helmets
- Crowded traffic is hard to monitor manually
- Small or blocked heads are difficult to detect
- Deep learning can support helmet monitoring



Project Scope

Clear focus helps explain the final system

PRIMARY OBJECTIVE

🪖 Helmet / No-Helmet Detection

✓ Detect helmet riders
✓ Detect no-helmet riders
✓ Identify helmet violations
✓ Evaluate detection performance

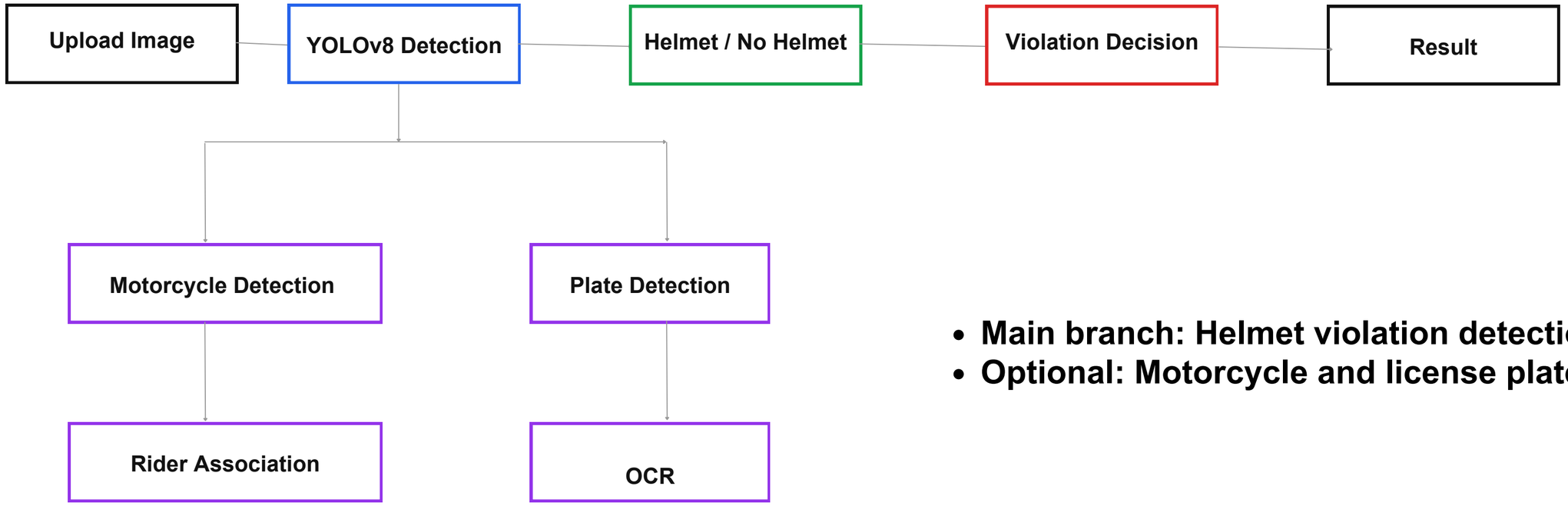
SUPPORTING FEATURES

🏍️ Motorcycle Detection

- Rider association
- 🔎 License Plate + OCR
- EasyOCR recognition
- FUTURE
- Improve these supporting features

System Workflow

From uploaded image to violation decision



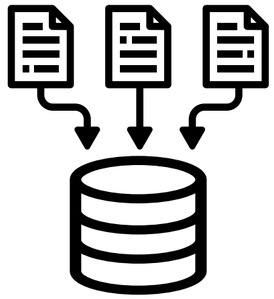
- **Main branch: Helmet violation detection**
- **Optional: Motorcycle and license plate features**

Dataset & Pipeline



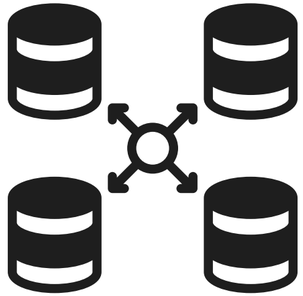
Dataset

8,481 images | Roboflow Version 4



Data Split

Train: 5,936 (70%)
Validation: 1,690 (20%)
Test: 855 (10%)



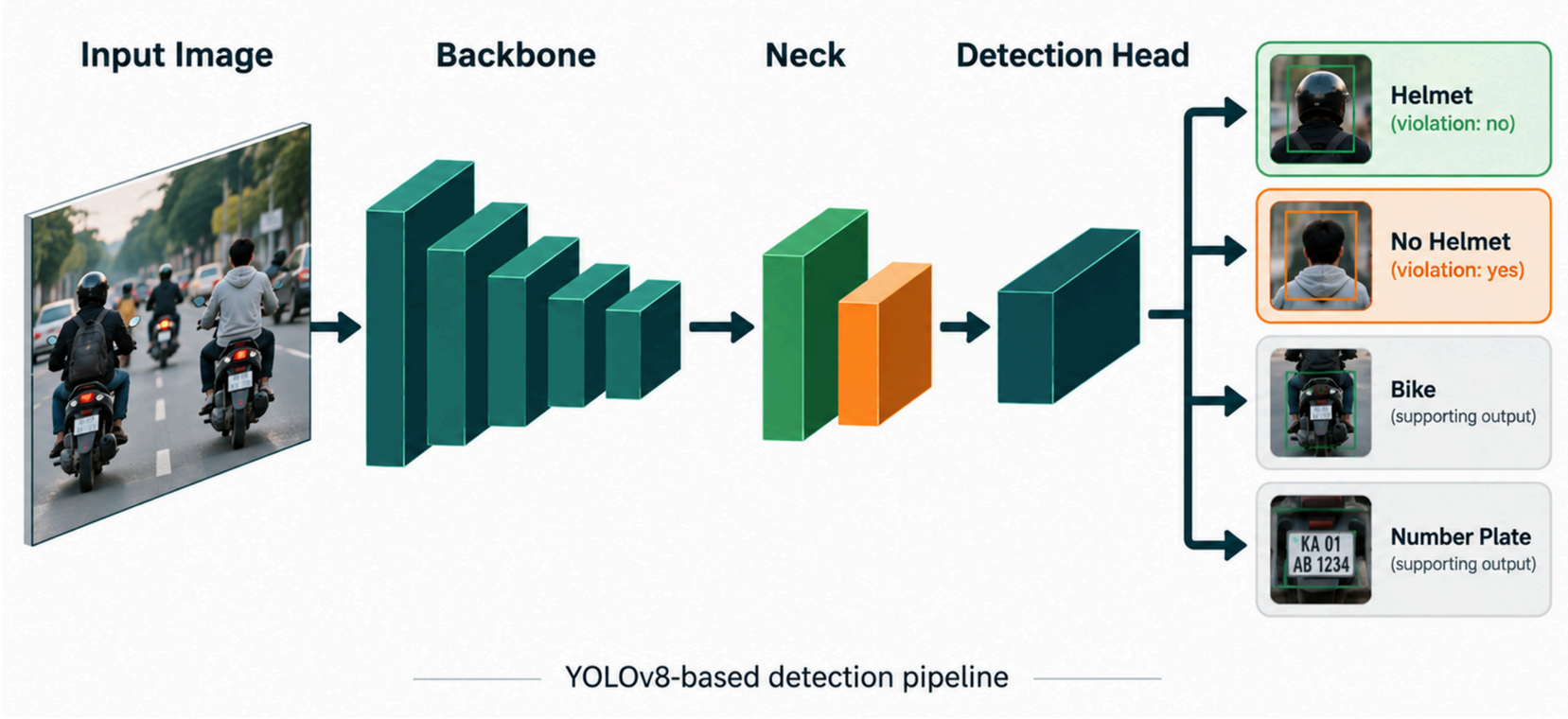
Classes

Bike — supporting
Helmet — main focus
No-Helmet — main focus
Number-Plate — optional

Preparation: Images → Labeling → Split → Resize 640×640 → YOLOv8 Training

Model Architecture & Training Strategy

YOLOv8 architecture, fine-tuning strategy, and tools



Training Losses

Box Loss • Classification Loss • DFL Loss

Model Architecture & Training Strategy

YOLOv8 architecture, fine-tuning strategy, and tools

TRAINING STRATEGY

Baseline Model
Original YOLOv8 trained on the dataset

Fine-Tuned Model
Continued training with difficult no-helmet examples

TOOLS



PyTorch + Ultralytics



Google Colab + Tesla T4



Gradio + EasyOCR

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Experimental Setup

Training configuration and evaluation environment

640 × 640	16	40	Tesla T4
Image size	Batch size	Epochs	GPU

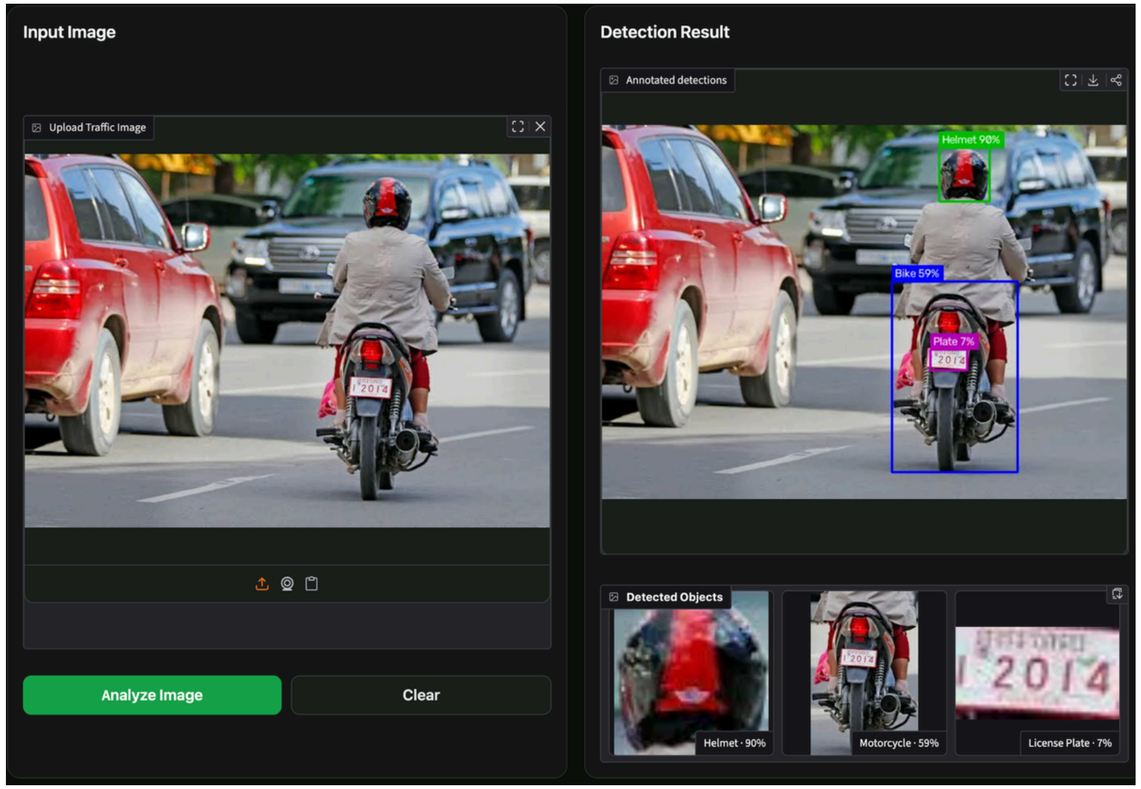
Dataset & Framework

- Train: 5,936 images
- Validation: 1,690 images
- Test: 855 images
- PyTorch + Ultralytics YOLOv8

Evaluation Metrics

- Precision • Recall
- mAP@50
- mAP@50:95

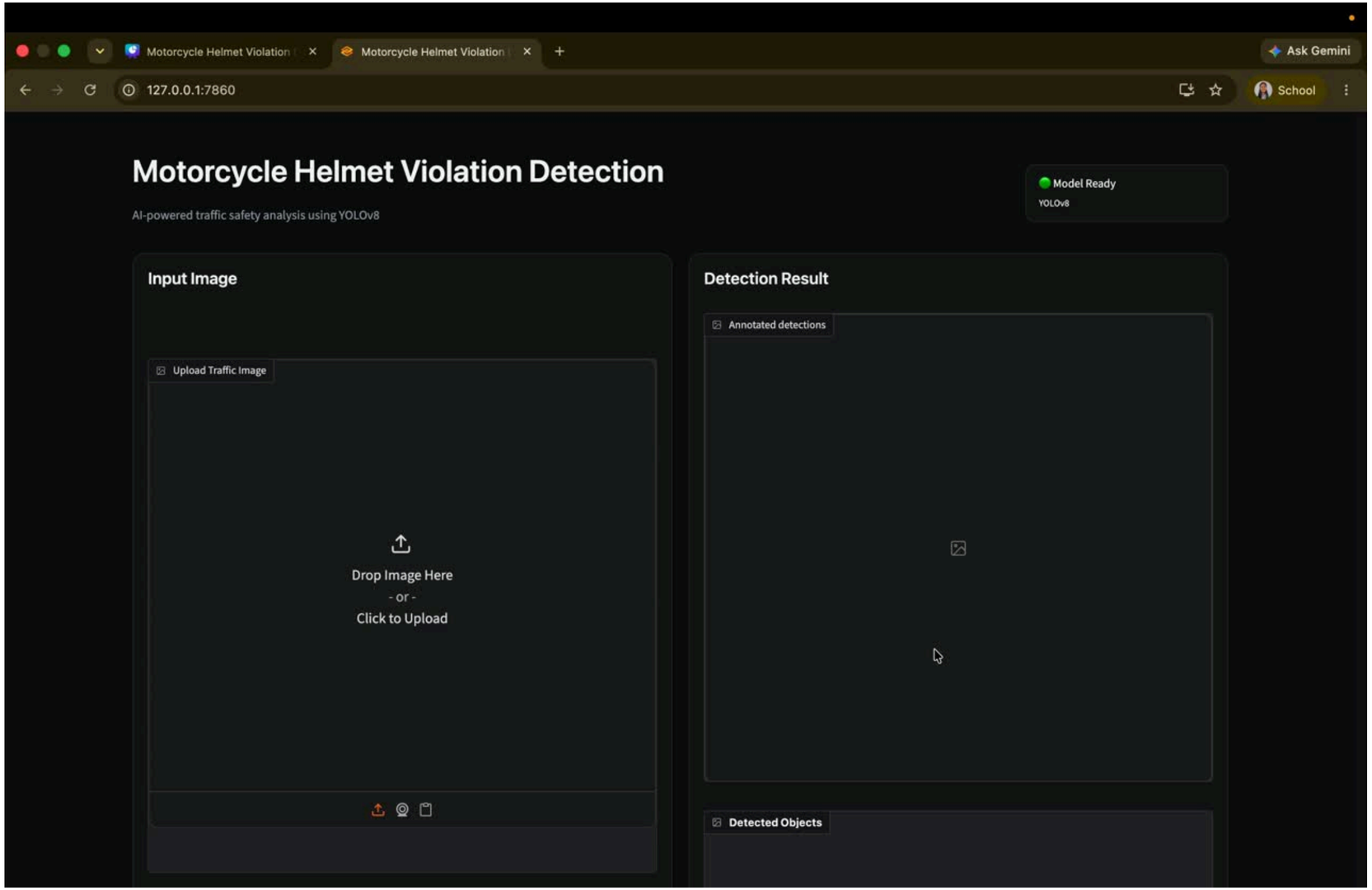
RESULTS & VISUAL COMPARISON




```
Classes: {0: 'bike', 1: 'helmet', 2: 'no-helmet', 3: 'number-plate'}
No-helmet class ID: 2
```

	Metric	Baseline	Fine-Tuned	Change
0	Precision	0.923	0.918	-0.005
1	Recall	0.881	0.888	0.007
2	mAP@50	0.942	0.943	0.001
3	mAP@50:95	0.669	0.691	0.022
4	No-Helmet Precision	0.912	0.906	-0.007
5	No-Helmet Recall	0.822	0.834	0.012
6	No-Helmet mAP@50	0.911	0.917	0.006
7	No-Helmet mAP@50:95	0.605	0.621	0.016

Demo


Motorcycle Helmet Violation
Motorcycle Helmet Violation
Ask Gemini
127.0.0.1:7860
School
Motorcycle Helmet Violation Detection
AI-powered traffic safety analysis using YOLOv8
Model Ready
YOLOv8
Input Image
Upload Traffic Image
Drop Image Here
- or -
Click to Upload
Detection Result
Annotated detections
Detected Objects

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Discussion & Error Analysis

Where the model still makes mistakes and what the results show

Common Errors

- Hats or caps can be confused with helmets
- Small or distant heads may be missed
- Occlusion and rear views reduce detection quality
- Multiple riders on one motorcycle are difficult

What the Results Show

- No-helmet recall improved after fine-tuning
- Difficult no-helmet cases are detected better
- Precision decreased slightly, so false detections remain
- Crowded scenes still cause missed or incorrect detections

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Conclusion & Future Work

What was achieved and what comes next

Conclusion

- Built a YOLOv8 system for helmet / no-helmet detection
- Fine-tuning improved difficult no-helmet cases
- Created a Gradio app for clear detection results

Future Work

- Add more difficult helmet / no-helmet training data
- Improve crowded scenes, multi-rider detection, bike and plate features
- Compare another detector and add video / live-camera support

Thank you